August 26, 2026

Search Committee Wallace H. Coulter Department of Biomedical Engineering Georgia Institute of Technology and Emory University via BMEchair@gatech.edu

Dear Members of the Search Committee,

I am writing to apply for the open-rank faculty position in the Wallace H. Coulter Department of Biomedical Engineering, as a candidate for a tenure-track Assistant Professor appointment. I am seeking to build a research program at the intersection of biomanufacturing quality science and neuropsychiatric/neurodegenerative disease modeling — a combination unusually well matched to the Coulter Department's strengths in cell manufacturing and neuroscience.

My proposed program develops the biomaterials, biological protocols, and computational infrastructure needed to make iPSC-derived neural and glial cell products reproducible, characterizable, and scalable, treating manufacturing rigor and disease modeling as one problem rather than two. Cell and gene therapy for neuropsychiatric and neurodegenerative disease is limited less by our ability to differentiate iPSCs into disease-relevant populations than by our ability to do so reproducibly, at scale, under real quality control. I propose to attack that bottleneck from both directions I know firsthand: characterizing iPSC-derived products against explicit critical quality attributes; using computational structural pharmacology and biomarker profiling to validate that those products model the disease biology they are meant to represent; and applying industrial-grade process control to their manufacture. Georgia Tech's NSF Engineering Research Center for Cell Manufacturing Technologies, the Marcus Center for Therapeutic Cell Characterization and Manufacturing, and the department's neuroscience community make this the strongest possible home for that vision.

My preparation is somewhat unconventional, and I consider that an asset rather than a gap to explain away. I hold a PhD in Biomedical Materials Science & Engineering from UTEP, where my dissertation developed a conductive carbon-nanotube–alginate hydrogel as a cell-therapy delivery vehicle. I then completed a postdoctoral appointment in neuropsychiatry at Texas Tech University Health Sciences Center El Paso, directing cGMP-compliant iPSC reprogramming and neural/glial differentiation and leading biobanking and transcriptomic analysis for neuropsychiatric disease research. Since 2024, I have managed a 60-plus person operations team at Amazon, driving an 82% throughput improvement and 99.7% quality compliance by building the process-control systems needed to make a large operation reproducible on demand — not a detour from research, but where I learned the discipline cell-therapy manufacturing most needs. My publication record includes seven peer-reviewed journal articles, spanning biomaterials characterization, tissue engineering, and neuropsychiatric transcriptomics, with one additional manuscript under review.

I take teaching and mentoring as a core part of this role. As a doctoral researcher I mentored more than thirty undergraduate and graduate students and taught a module of a biomaterials characterization course; at Amazon I built a training pipeline that has since developed both frontline and management-level staff. Georgia Tech's stated commitment to student-first mentorship, collaboration across difference, and rigorous, accountable research matches how I have already tried to work across a career spanning materials science, clinical regulatory practice, neuropsychiatric research, and industrial operations.

Enclosed are my curriculum vitae, statement of research interests, statement of teaching interests, statement on Georgia Tech's mission and values, and reference contact information. I would welcome the opportunity to discuss this position further and am available at your convenience.

Sincerely,

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